Data Structures Group Project | Artificial Intelligence Usage Disclosure

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IMPORTANT

AI Disclosure: This project utilizes AI-assisted coding tools. A full disclosure of its use, as required by university policy, can be found in AI Disclosure.pdf.

A full-featured desktop E-Commerce application built with **JavaFX** and **PostgreSQL**, engineered with custom data structures and design patterns. Repository can be found in BrandEx Repo.

Project Overview

BrandEx is a desktop-based E-Commerce platform for browsing, purchasing, and managing products. It provides two distinct experiences:

- **Customer Portal**: Browse products, manage a cart, place orders, and view your profile.
- **Admin Dashboard**: Manage products, users, and order fulfillment from a central control panel.

Technical Architecture

Stack

Layer	**Technology**
Language	Java 25
UI Framework	JavaFX 25
UI Theme	AtlantaFX (Cupertino Dark)
Database	PostgreSQL
Build Tool	Apache Maven
Email	Jakarta Mail

Packaging	`Jpackage` (Windows EXE/MSI)

Package Structure

```
com.brandex/
├── App.java                # JavaFX Application entry point
├── Main.java               # Launcher entry point (jpackage safe)
├── command/                # Command Pattern (Undo/Redo cart
actions)
│   ├── Command.java
│   ├── CartAddCommand.java
│   └── CartRemoveCommand.java
├── database/
│   ├── JDBC.java           # Centralized prepared statement executor
│   └── DatabaseException.java
├── datastructures/         # Custom implementations
│   ├── BST.java            # Binary Search Tree
│   ├── LinkedList.java     # Linked List
│   ├── Queue.java          # Queue (Order processing)
│   ├── Stack.java          # Stack (Undo/Redo)
│   └── Node.java
├── models/                 # ADT Models
│   ├── Model.java          # Generic parent model
│   ├── Product.java
│   ├── User.java
│   ├── Cart.java, CartItem.java
│   ├── Order.java, OrderItem.java
│   └── enums/
│       ├── OrderStatus.java
│       ├── PaymentMethod.java
│       └── UserStatus.java
├── repository/             # Database Access Objects (CRUD)
│   ├── ProductRepository.java
│   ├── UserRepository.java
│   ├── CartRepository.java
```

```
│    └── OrderRepository.java
├── service/                # Business logic layer
│    ├── AuthService.java
│    ├── ProductService.java
│    ├── CartService.java
│    ├── OrderService.java
│    └── UserService.java
├── ui/                     # JavaFX Controllers
│    ├── AuthController.java
│    ├── DashboardController.java
│    ├── ProductCatalogController.java
│    ├── CartController.java
│    ├── ProfileController.java
│    ├── ProfileFormController.java
│    ├── ProfilePasswordFormController.java
│    ├── CheckoutController.java
│    ├── OrderHistoryController.java
│    ├── ManageProductsController.java
│    ├── ManageOrdersController.java
│    ├── ManageUsersController.java
│    ├── ProductFormController.java
│    └── EditUserDialogController.java
└── utilities/
     ├── ConfigLoader.java
     ├── EmailSender.java
     ├── ImageLoader.java
     ├── InputValidator.java
     ├── OTPGenerator.java
     ├── PasswordHasher.java
     └── StatusLabelHelper.java
```

Design Patterns in Use

Pattern	Where Used

Singleton	All Services and Repositories
Repository	`ProductRepository`, `UserRepository`, `CartRepository`, `OrderRepository`
Service Layer	`AuthService`, `ProductService`, `CartService`, `OrderService`, `UserService`
Command Pattern	Cart add/remove with full undo/redo support
BST (Binary Search Tree)	In-memory product and user stores; efficient sorted traversal
Linked List	Cart item storage
Stack	Undo/Redo command history

System Requirements

End User (Running the Installer)

- **OS**: Windows 10 or later (64-bit)
- **Memory**: 4 GB RAM minimum (8 GB recommended)
- **Storage**: 300 MB free disk space

- **Network**: Internet connection for product images and email notifications.

Developer (Building from Source)

- **JDK**: Java 25+
- **Build Tool**: Apache Maven 3.8+
- **Database**: PostgreSQL 14+
- **IDE**: IntelliJ IDEA, VS Code, or Eclipse (with Java extensions)

Setup & Installation

Database Setup

1. **Install PostgreSQL** and create a new database:

```
CREATE DATABASE brandex;
```

2. **Run the schema script** to create all required tables:

```
-- Users table
   CREATE TABLE users (
     id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
     username TEXT UNIQUE NOT NULL,
     email TEXT UNIQUE NOT NULL,
     first_name TEXT,
     last_name TEXT,
     phone_number TEXT,
     shipping_address TEXT,
     role TEXT DEFAULT 'customer',
     status TEXT DEFAULT 'ACTIVE',
     password_hash TEXT NOT NULL,
     prev_hash_1 TEXT,
     prev_hash_2 TEXT,
     otp_hash TEXT,
     otp_used BOOLEAN DEFAULT FALSE,
     force_pw_change BOOLEAN DEFAULT FALSE,
```

```sql
  profile_image_url TEXT,
  created_at TIMESTAMPTZ DEFAULT NOW()
);

-- Products table
CREATE TABLE products (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name TEXT NOT NULL,
  description TEXT,
  category TEXT,
  brand TEXT,
  image_url TEXT,
  price NUMERIC(10, 2) DEFAULT 0.00,
  rating NUMERIC(3, 1) DEFAULT 0.0,
  stock INT DEFAULT 0,
  created_at TIMESTAMPTZ DEFAULT NOW()
);

-- Carts table
CREATE TABLE carts (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID REFERENCES users(id) ON DELETE CASCADE,
  total_price NUMERIC(10, 2) DEFAULT 0.00,
  created_at TIMESTAMPTZ DEFAULT NOW(),
  updated_at TIMESTAMPTZ DEFAULT NOW()
);

-- Cart items table
CREATE TABLE cart_item (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  cart_id UUID REFERENCES carts(id) ON DELETE CASCADE,
  product_id UUID REFERENCES products(id) ON DELETE CASCADE,
  quantity INT DEFAULT 1,
  total_price NUMERIC(10, 2) DEFAULT 0.00,
  created_at TIMESTAMPTZ DEFAULT NOW()
);

-- Orders table
CREATE TABLE orders (
```

```sql
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID REFERENCES users(id),
    order_number TEXT UNIQUE,
    status TEXT DEFAULT 'PENDING',
    shipping_address TEXT,
    payment_method TEXT,
    total_price NUMERIC(10, 2) DEFAULT 0.00,
    created_at TIMESTAMPTZ DEFAULT NOW()
  );

  -- Order items table
  CREATE TABLE order_item (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    order_id UUID REFERENCES orders(id) ON DELETE CASCADE,
    product_id UUID REFERENCES products(id),
    quantity INT DEFAULT 1,
    total_price NUMERIC(10, 2) DEFAULT 0.00,
    created_at TIMESTAMPTZ DEFAULT NOW()
  );
```

Configuration

The application reads from a `config.properties` file located at:

```
src/main/resources/com/brandex/config.properties
```

Create this file with the following keys:

```properties
# Database Connection
db.URL=jdbc:postgresql://localhost:5432/brandex
db.USER=YOUR_DB_USERNAME
db.PASSWORD=YOUR_DB_PASSWORD

# Product Images Base URL (where your product images are hosted)
db.PRODUCT_IMAGE_PATH=https://your-image-host.com/products/
```

```
# Email (SMTP via Gmail -- requires an App Password)
email.user=your-email@gmail.com
email.password=your-gmail-app-password
```

NOTE

Configuration File: The `config.properties` file containing the environment-specific database and email credentials **will be provided separately during the project submission**. Please ensure this file is placed in `src/main/resources/com/brandex/` before attempting to build or run the source code.

Gmail App Passwords: To use Gmail for sending OTP and notification emails, enable 2-Factor Authentication on your Google account, then create an "App Password" under your Google Account Security settings. Use that 16-character password in `mail.password`.

Running the Application

Option 1 — Via Maven (Development)

```
mvn clean javafx:run
```

IMPORTANT

Prerequisites for Maven: You must have **Apache Maven** installed on your system to run the `mvn clean javafx:run` command. Alternatively, if you are using **VS Code**, you can install the **Extension Pack for Java** and run the application directly from the editor by clicking "Run Java" above the `main` method in `App.java`.

Installation Links:

- ❖ Download Java JDK 25+
- ❖ Download Apache Maven

Option 2 — Via the Windows Installer (End Users)

1. Download the `BrandEx-1.0.exe` from the Releases page.

2. Run the installer.

NOTE

Security Note: Because this is an educational project, the installer is not digitally signed. You may see a Windows SmartScreen warning stating "Windows protects your PC." To proceed, click **"More info"** and then **"Run anyway"**.

3. Follow the prompts to complete installation.

4. Launch **BrandEx** from the Desktop or Start Menu shortcut.

Building the Installer

To build the Windows installer from source:

```
# Step 1: Package the app and copy all dependencies
mvn clean package dependency:copy-dependencies

# Step 2: Copy the main JAR into the dependency folder
cp target/brandex-1.0-SNAPSHOT.jar target/dependency/

# Step 3: Run jpackage to create the .exe installer
jpackage \
  --input target/dependency \
  --name BrandEx \
  --main-jar brandex-1.0-SNAPSHOT.jar \
```

```
  --main-class com.brandex.Main \
  --type exe \
  --app-version 1.0 \
  --vendor "BrandEx Group" \
  --icon src/main/resources/com/brandex/images/icons/logo.ico \
  --win-shortcut \
  --win-menu \
  --win-dir-chooser \
  --win-shortcut-prompt \
  --win-menu-group "BrandEx"
```

The resulting `BrandEx-1.0.exe` will appear in your project root.

Customer Features

Account Management

Registering an Account

1. Launch BrandEx, the Login/Register screen appears.

2. Click the **Register** tab.

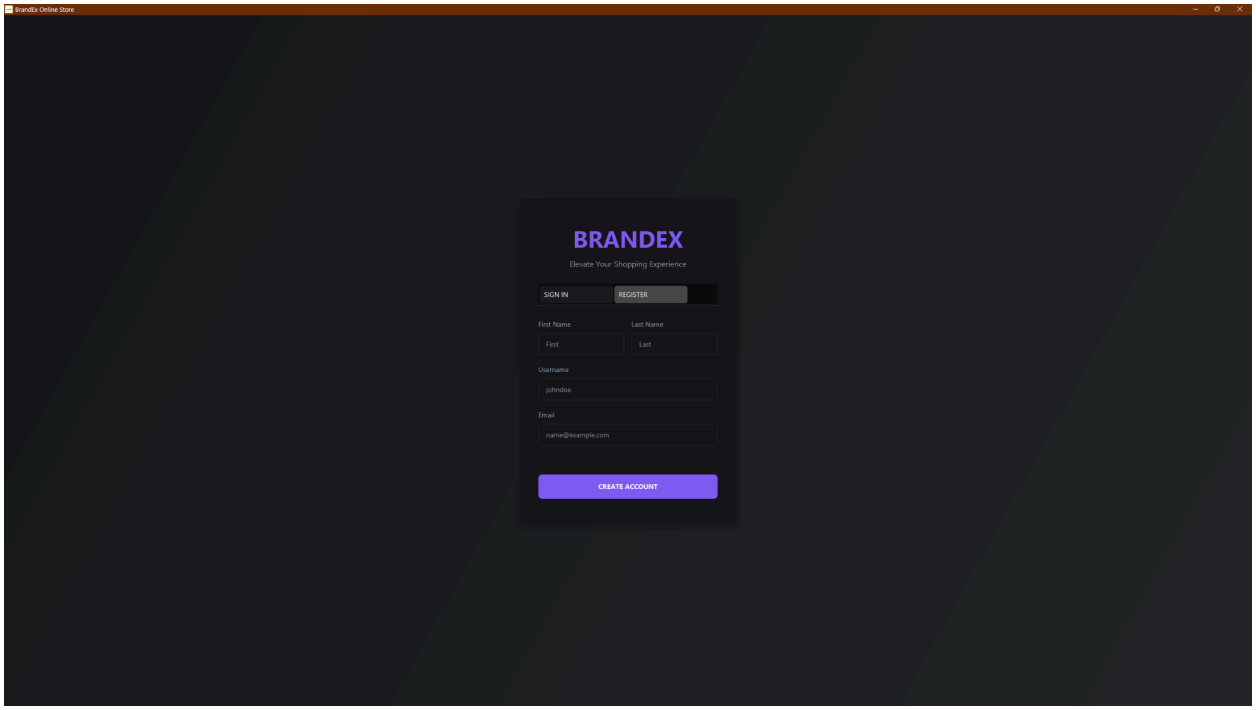
3. Enter your **First Name**, **Last Name**, **Email**, and **Username**.

4. Click **Register**.

5. A **One-Time Password (OTP)** will be sent to your email address.

6. On the OTP screen, enter the code from your email.

7. You will be prompted to **set a permanent password** on first login.

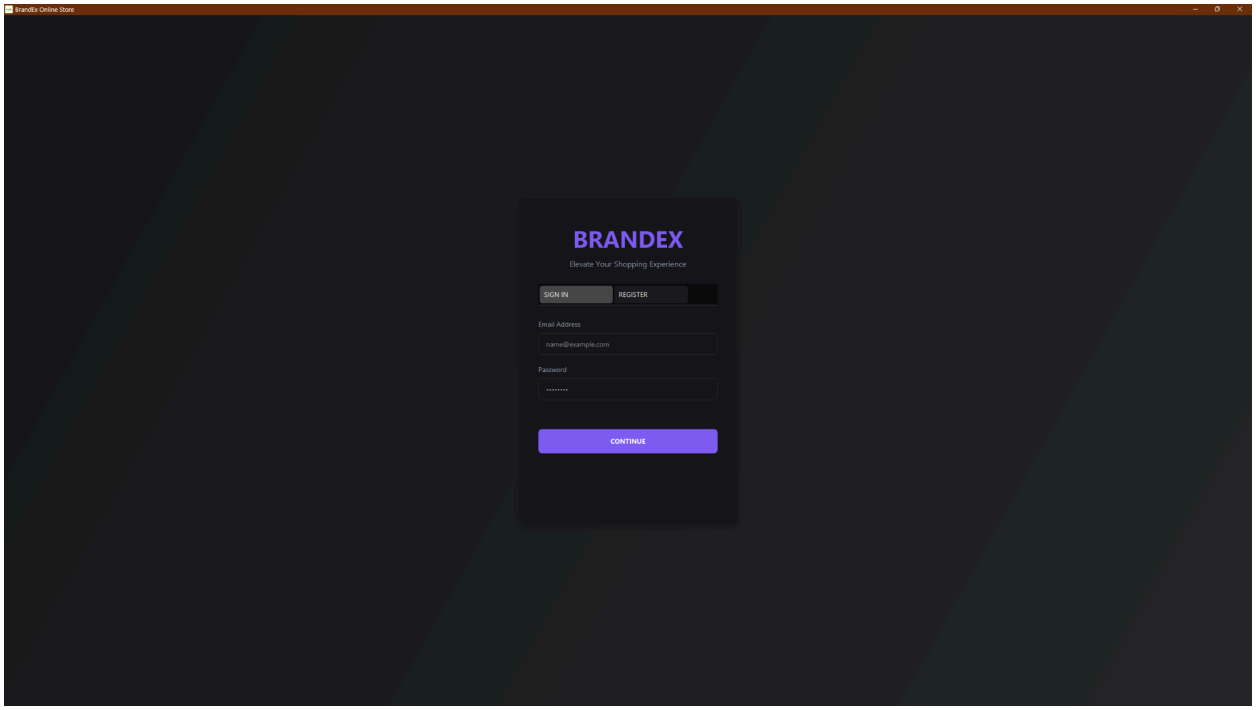


Logging In

1. On the Login tab, enter your registered **Email** and **Password**.

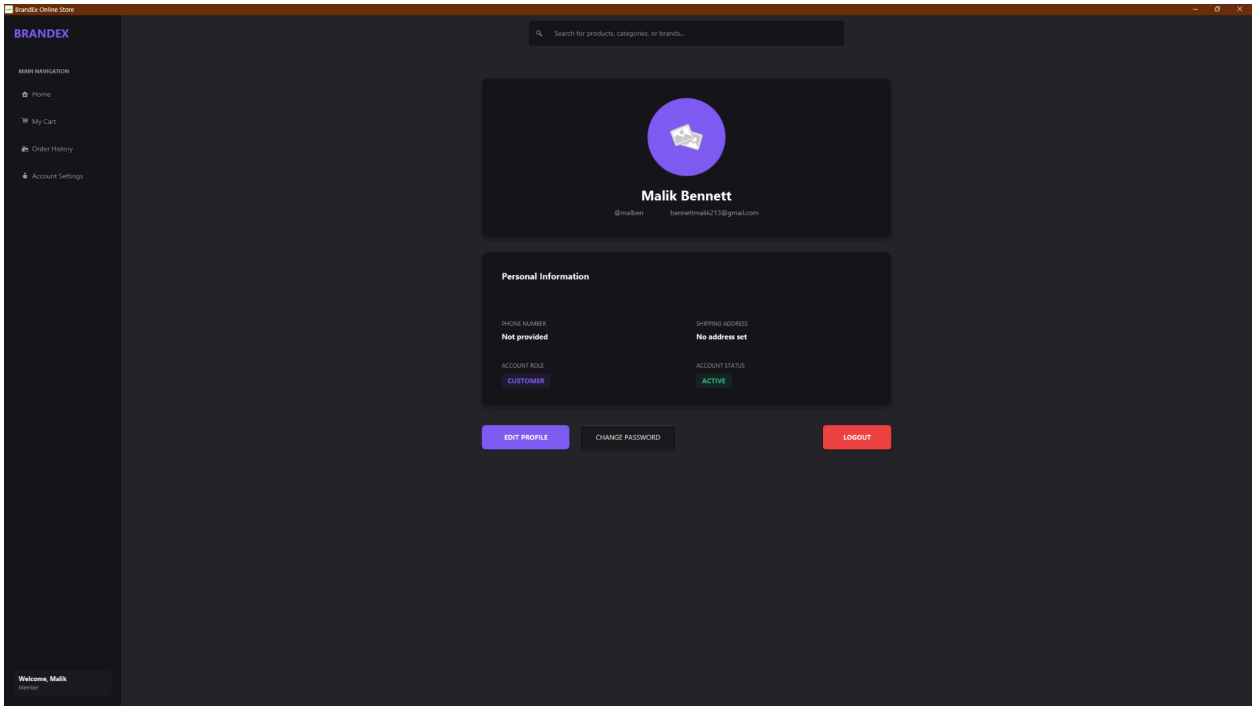
2. Click **Login**.

3. If your password has been reset by an admin, you will be required to change it before proceeding.



Viewing & Editing Your Profile

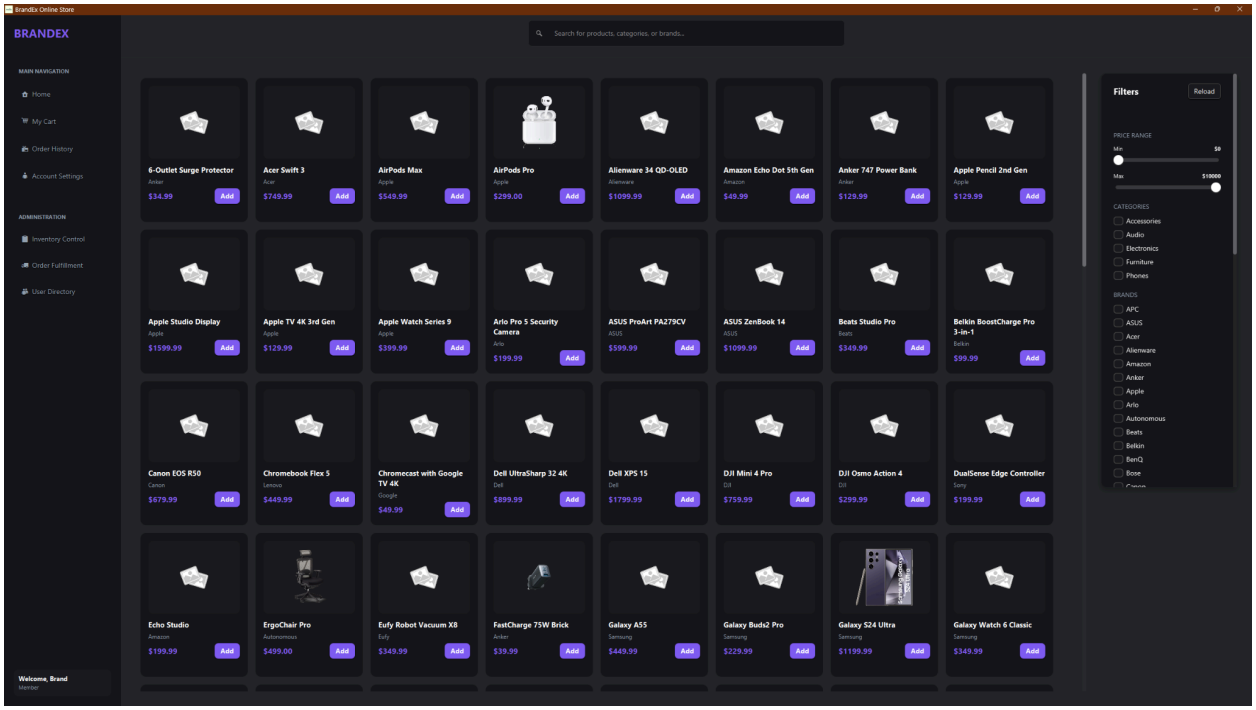
1. Once logged in, click the **Profile** button in the top navigation bar.
2. Your profile page displays:
    - ❖ Full Name, Username, and Email
    - ❖ Phone Number and Shipping Address
    - ❖ Account Role and Status (badge)
3. Click **Edit Profile** to update your information.
4. Click **Change Password** to update your credentials.
5. Click **Logout** to securely end your session.



Shopping

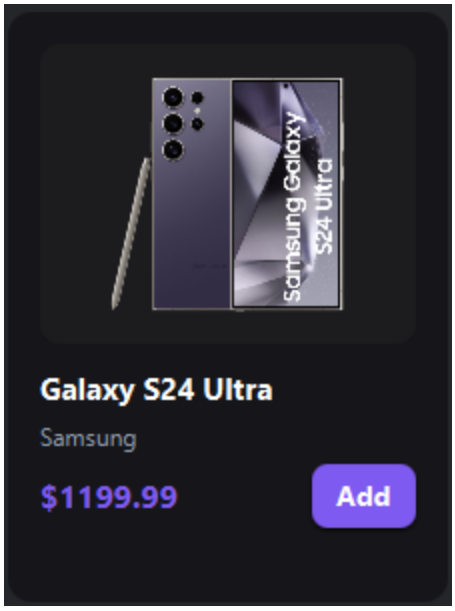
Browsing Products

1. After logging in, the **Product Catalog** is displayed by default.
2. Products are shown as cards with image, name, brand, price, rating, and stock.
3. Use the **Search Bar** at the top to filter products by name in real time.
4. Clear the search bar and press Enter to show all products again.



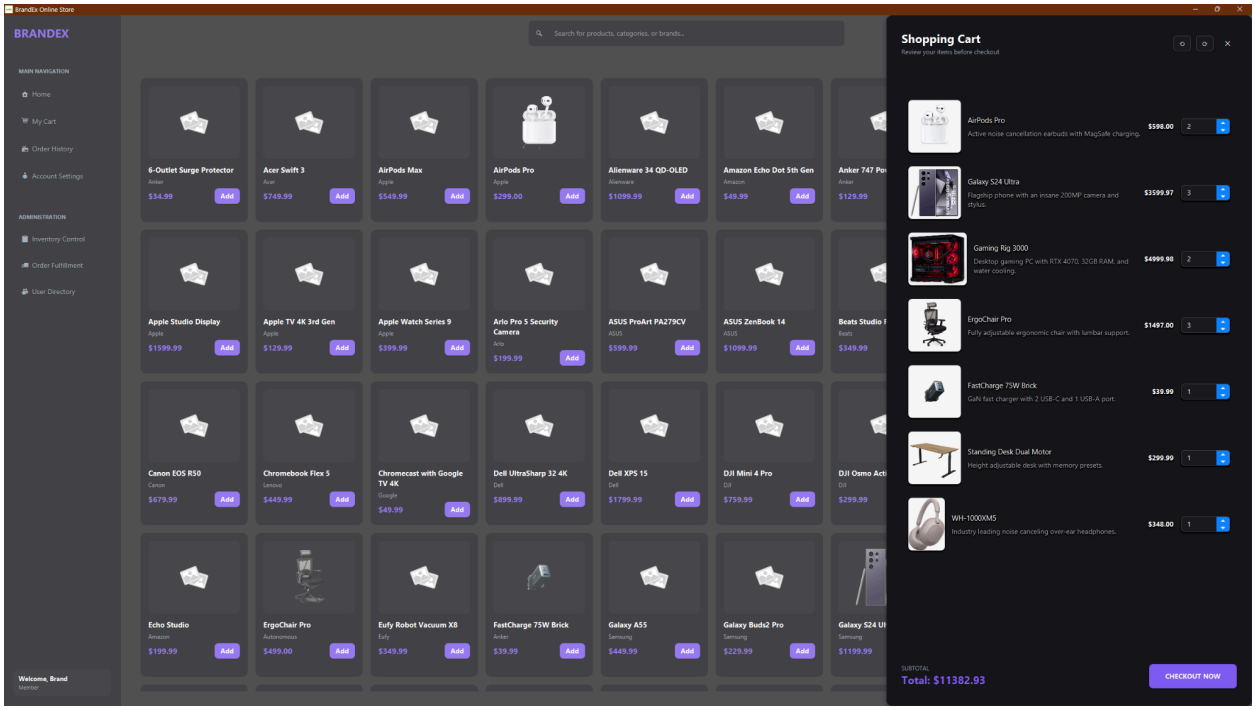
Viewing Product Details

1. Each product card shows:
    - ❖ Product Name & Brand
    - ❖ Price and current Stock
    - ❖ Star Rating
    - ❖ Product Image
2. On any product card, click the **Add to Cart** button.
3. You can add the same item multiple times to increase quantity.
4. The item is immediately added to your session cart.



Cart Management

1. Click the **Cart** icon/button in the navigation bar to open the cart panel.
2. The cart displays:
    - Each item with its name and total price
    - The running **Cart Total**
3. To **remove an item**, click the remove or delete button next to it.
4. Use **Undo** to restore a removed item.
5. Use **Redo** to re-apply a change.
6. Click **Checkout** to proceed to the payment screen.



Checkout & Orders

Placing an Order

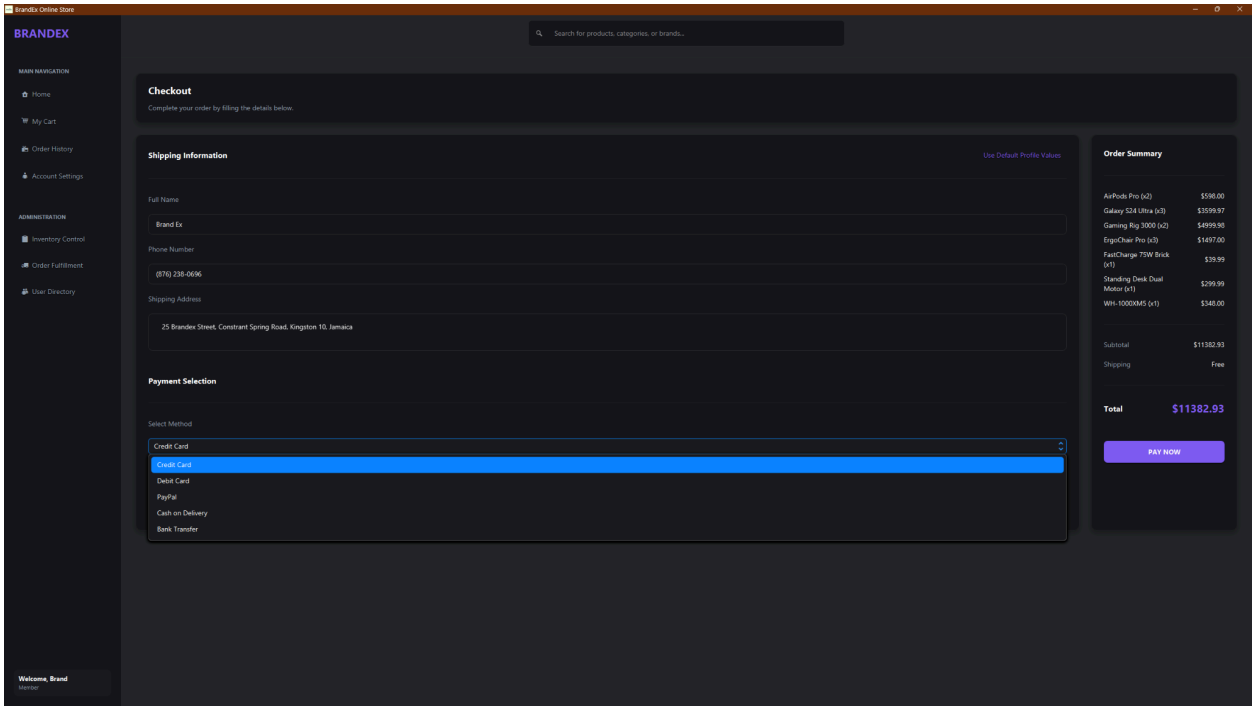
1. From the Cart, click **Checkout**.

2. Review your cart summary and total.

3. Confirm your **Shipping Address**.

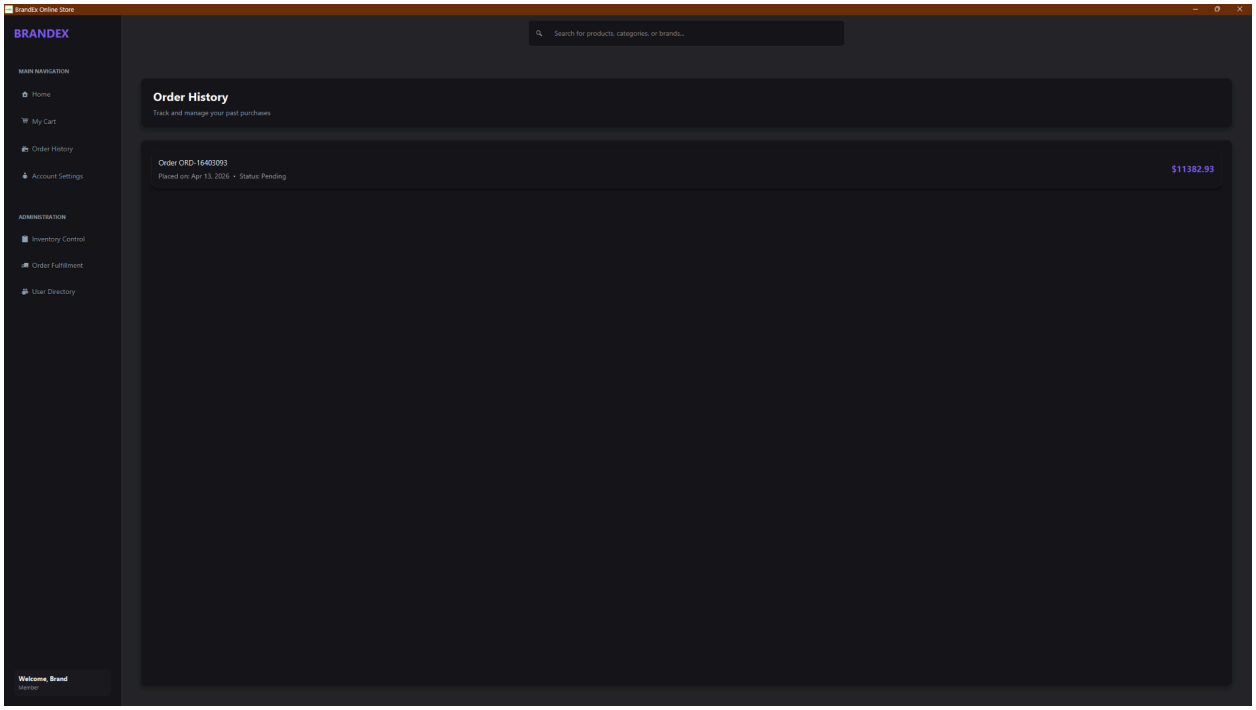
4. Select a **Payment Method**.

5. Click **Place Order** to finalize.



Viewing Order History

1. Click **Orders** in the navigation bar.
2. A list of your past orders is shown with:
    - ❖ Order Number
    - ❖ Date placed
    - ❖ Status (Pending, Shipped, Delivered)
    - ❖ Order Total



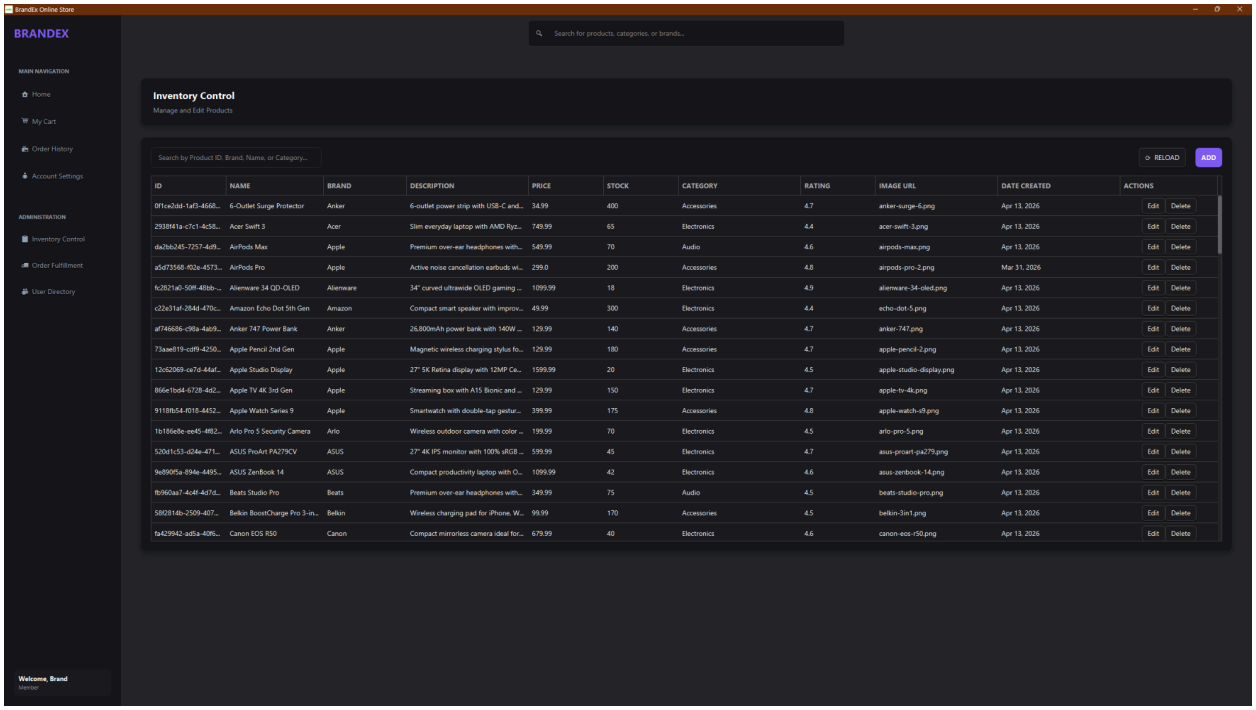
Admin Features

Admin accounts have access to an extended navigation bar with management tools. Admin access is determined by the `role` field on the user account.

Product Management

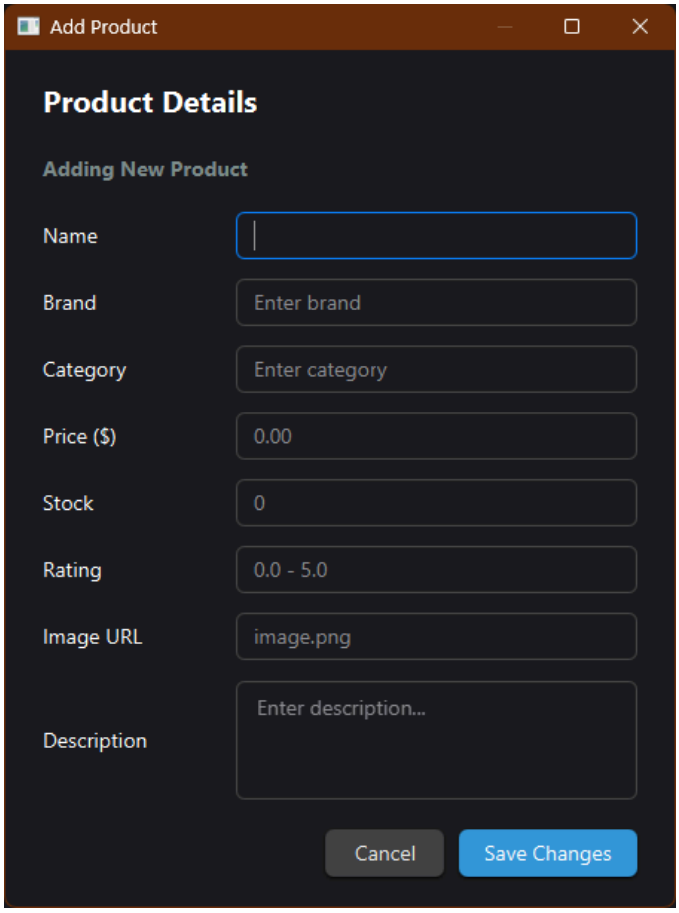
Viewing All Products

- A sortable table lists all products in the system.
- Columns: ID, Name, Brand, Description, Price, Stock, Category, Rating, Image URL, Date Created, Actions.



Adding a New Product

1. Click the **Add Product** button (top-right of the table).
2. A form dialog opens with empty fields.
3. Fill in: Name (required), Brand, Category, Price (required), Stock (required), Rating, Image URL, and Description.
4. Click **Save** to create the product. It is immediately added to the catalog.
5. Click **Cancel** to discard changes.



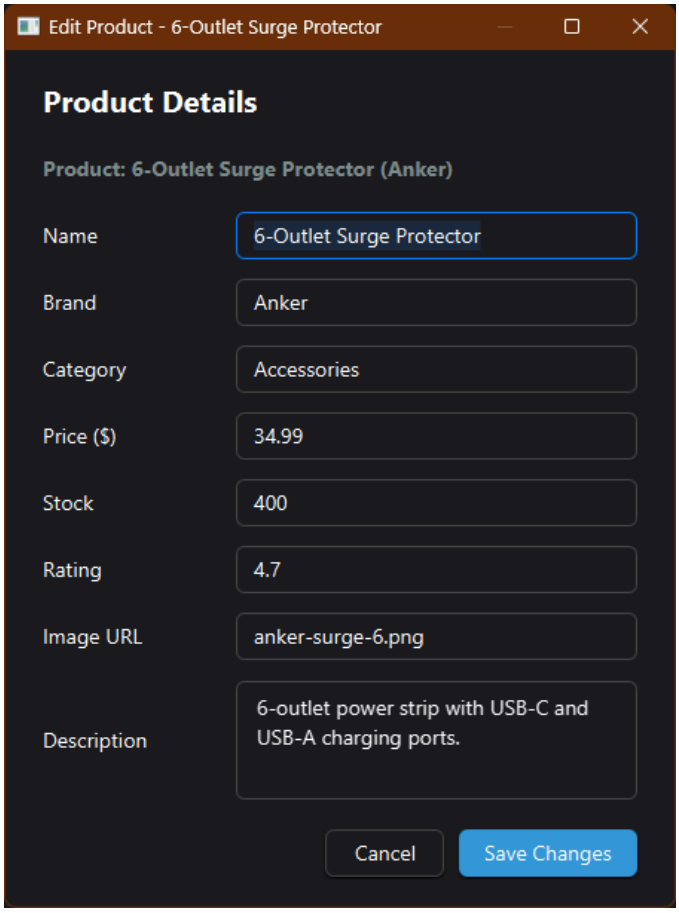
Editing a Product

1. In the product table, click the **Edit** button on the desired row.

2. The same form dialog opens, pre-populated with the product's current data.

3. Modify any fields.

4. Click **Save**. The product is updated in the database and the in-memory catalog simultaneously.

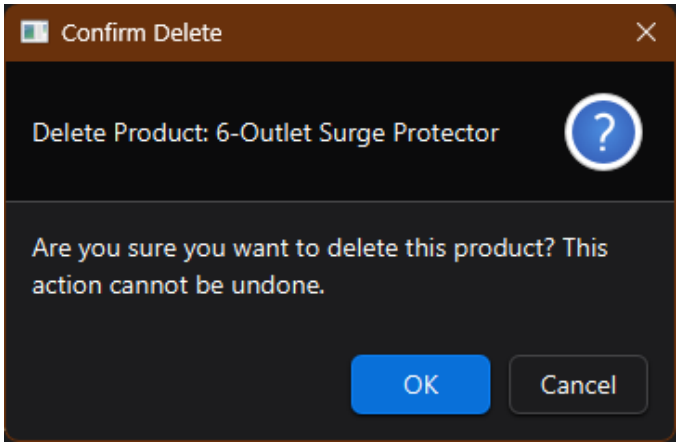


Deleting a Product

1. Click the **Delete** button on the desired product row.

2. A confirmation dialog appears: *"Are you sure you want to delete this product? This action cannot be undone."*

3. Click **OK** to permanently delete. The product is removed from both the database and the live catalog.

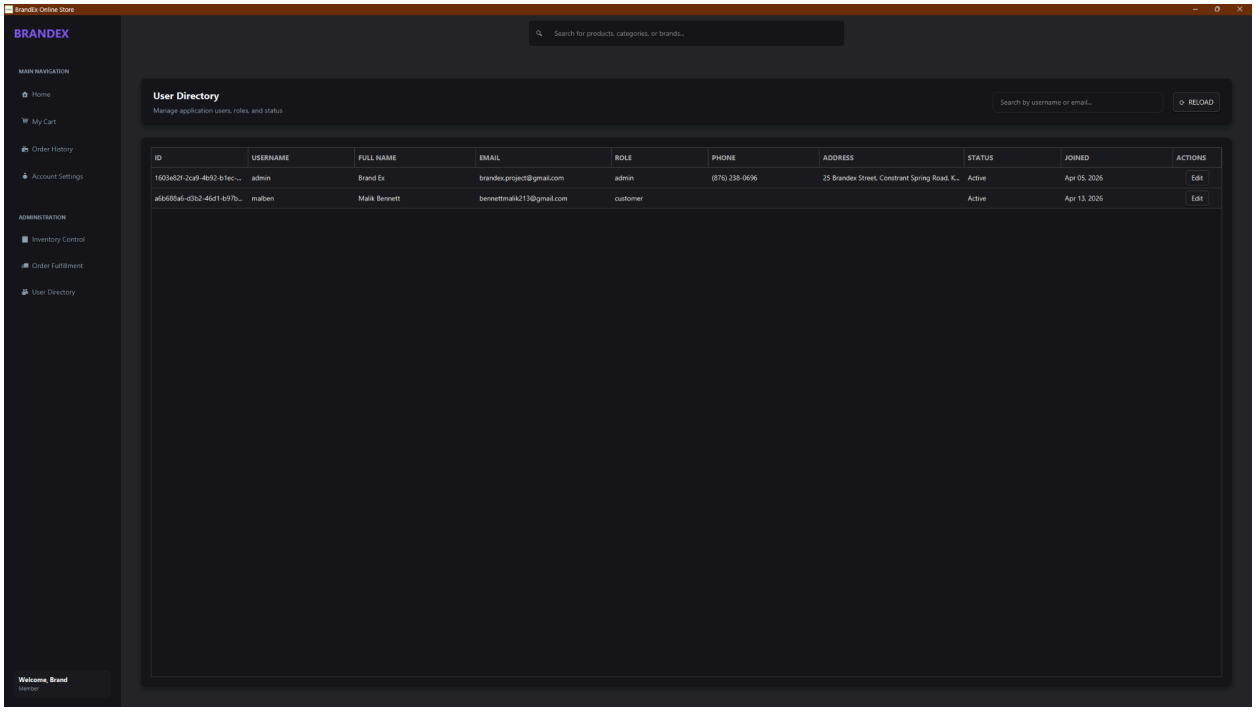


User Management

Navigate to: **Admin Nav → Manage Users**

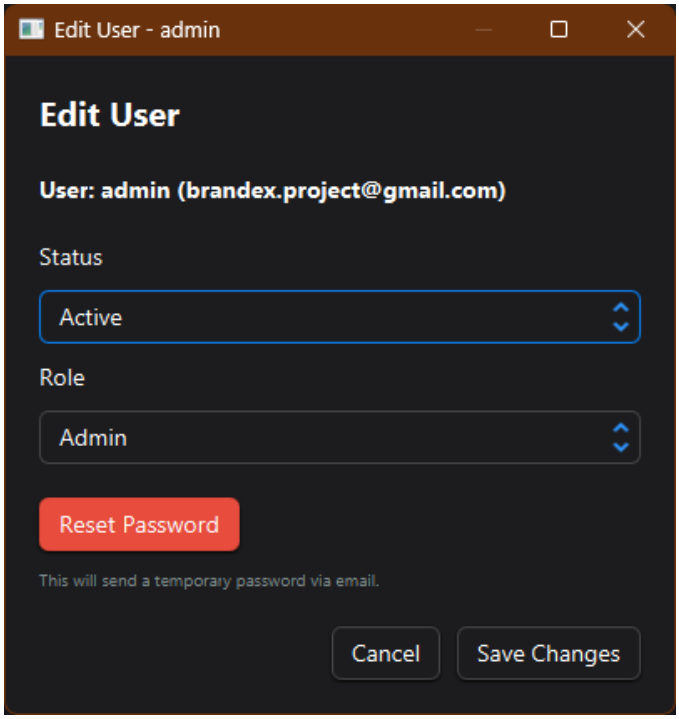
Viewing All Users

- A table lists all registered customer accounts.
- Columns: Full Name, Email, Username, Status, Role, Phone, Address, Joined Date, Actions.



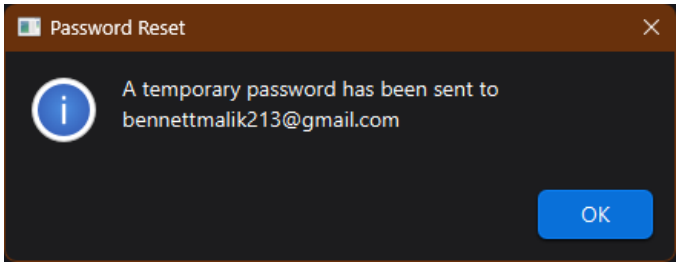
Editing a User's Status and Role

1. Click the **Edit** button on any user row.
2. An **Edit User Dialog** opens.
3. Modify the user's **Status** (Active, Suspended, Banned) and/or **Role** (customer, admin).
4. Click **Save** to apply changes.



Resetting a User's Password

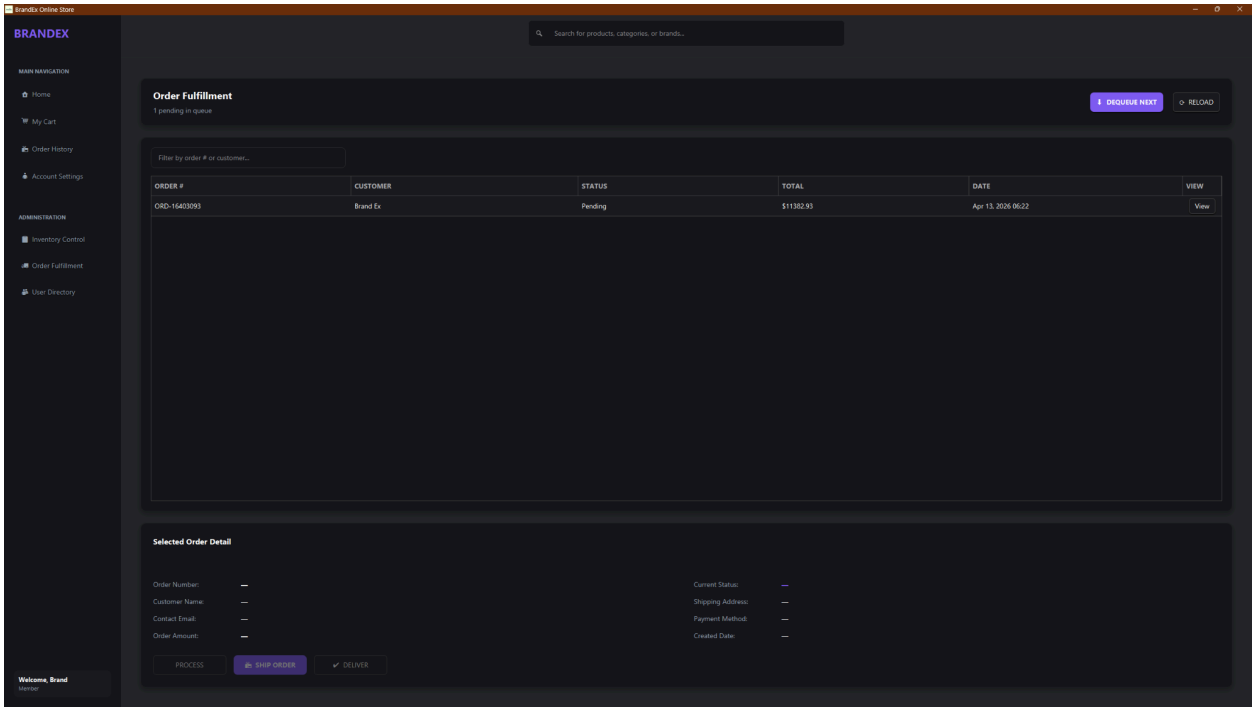
1. Click the **Reset Password** button in the Edit User Dialog.
2. A temporary password is automatically generated and emailed to the user.
3. The user's account is flagged with `force_pw_change = true`, requiring them to set a new password on the next login.



Order Processing

Navigate to: **Admin Nav → Manage Orders**

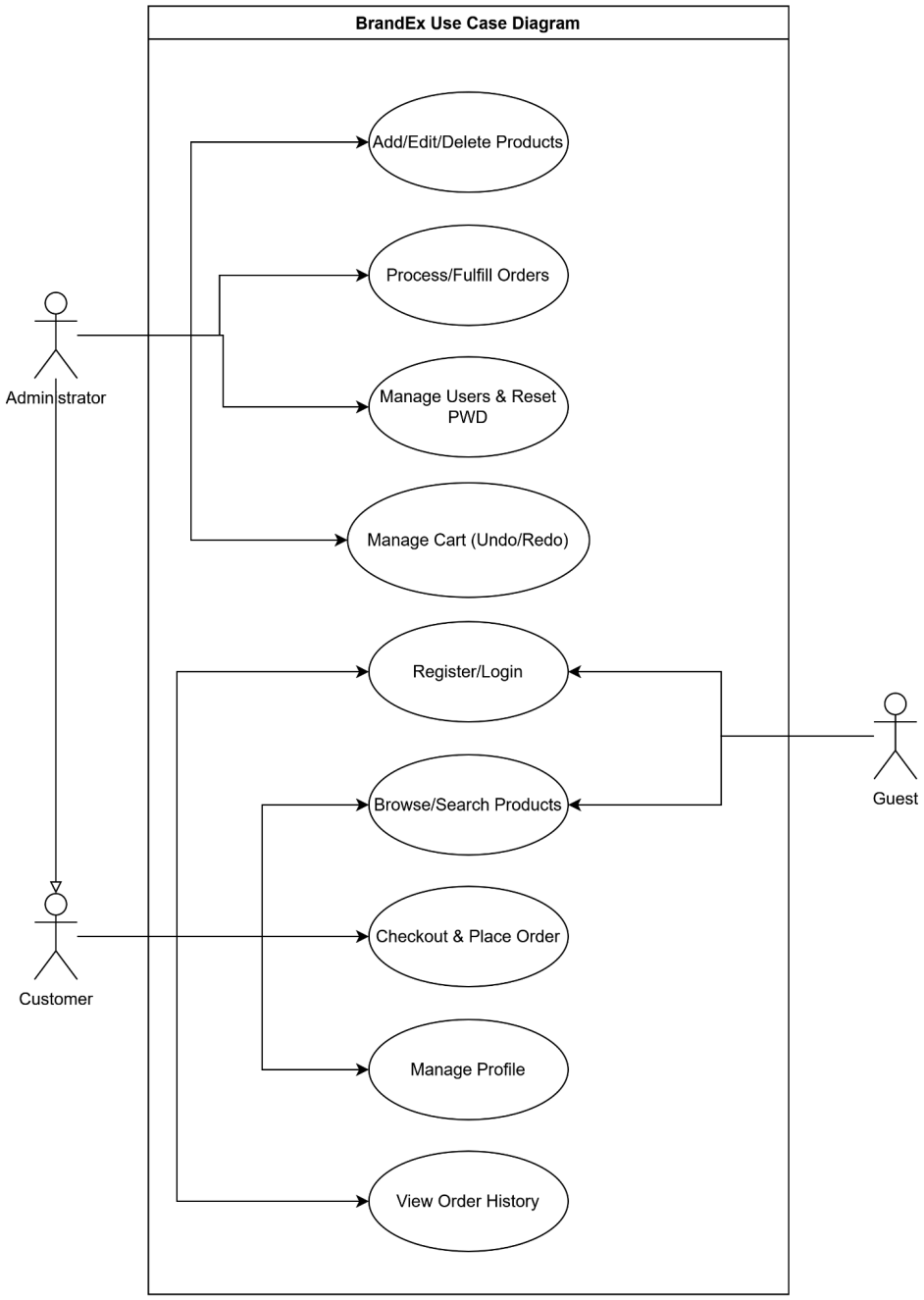
- View all pending orders in a queue.
- Process and update order statuses (Pending → Shipped → Delivered).
- Send customer notifications upon status changes.



UML Diagrams

Use Case Diagram

The Use Case Diagram illustrates the interactions between system actors (Customer, Admin) and the available system features.



- **Guest**: Can view the login/register screen.
- **Customer**: Can browse products, manage cart, place orders, and view their profile.
- **Administrator**: Has all Customer capabilities plus product CRUD, user management, and order processing.

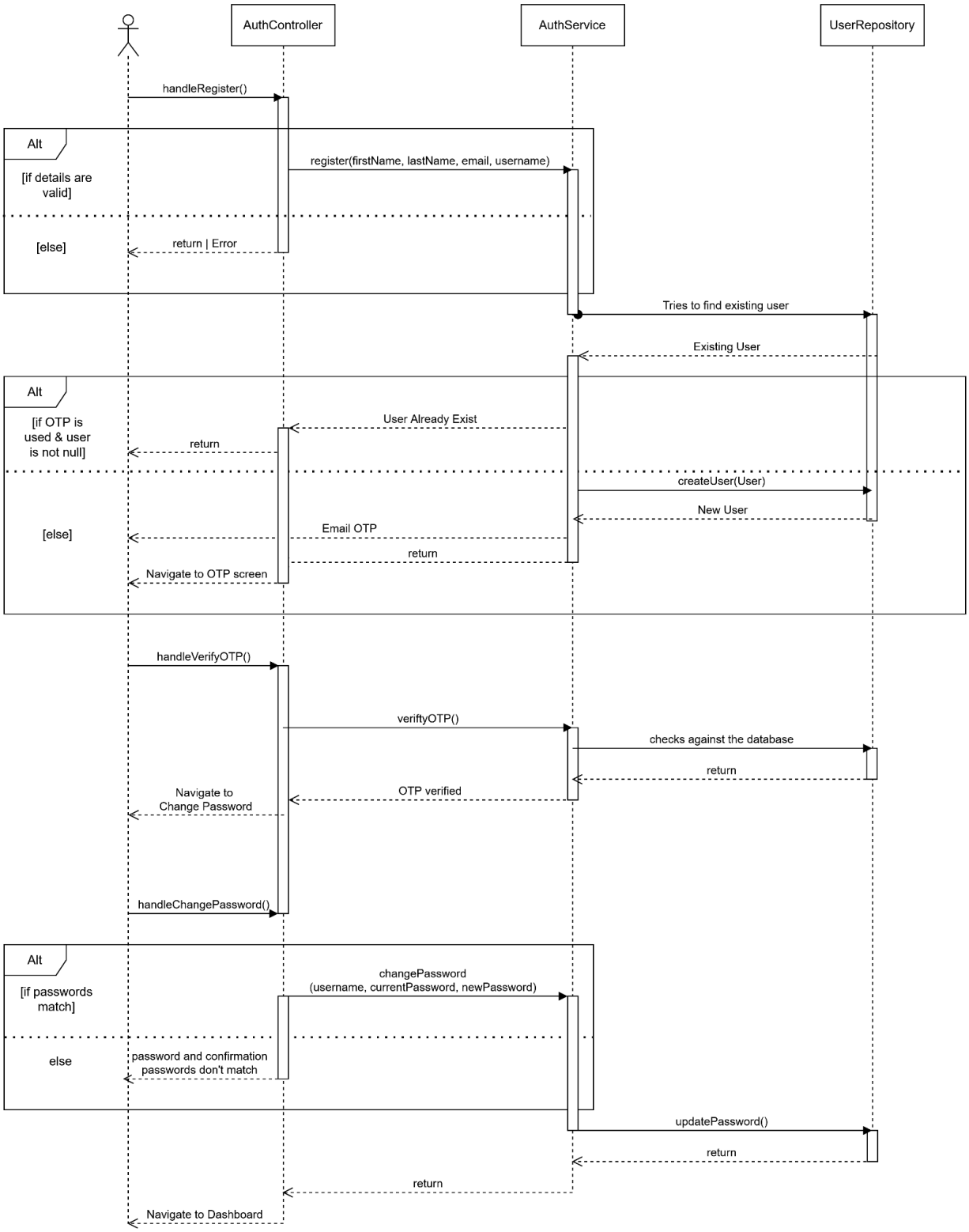
Class Diagram

The Class Diagram illustrates the domain model, technical layers, and the relationships between key classes.

View Full Detailed Class Diagram (PDF)

Sequence & Activity Diagrams

Illustrates the flow of messages through the system and the core business logic for key workflows like Registration.


AuthController
AuthService
UserRepository
handleRegister()
Alt
[if details are valid]
register(firstName, lastName, email, username)
[else]
return | Error
Tries to find existing user
Existing User
Alt
[if OTP is used & user is not null]
User Already Exist
return
createUser(User)
New User
[else]
Email OTP
return
Navigate to OTP screen
handleVerifyOTP()
veriftyOTP()
checks against the database
return
OTP verified
Navigate to Change Password
handleChangePassword()
Alt
[if passwords match]
changePassword (username, currentPassword, newPassword)
else
password and confirmation passwords don't match
updatePassword()
return
return
Navigate to Dashboard

Common Setup Issues

Problem	Likely Cause	Solution
`Error loading FXML` on startup	Missing or corrupt resource files	Ensure you built from source with `mvn clean package`
`Connection refused` (DB error)	PostgreSQL not running	Start your PostgreSQL service and verify credentials in `config.properties`
Application opens but is blank	FXML controller binding issue	Check console for a `NullPointerException`, ensure all `fx:id` values are correct
OTP email not received	SMTP credentials invalid	Regenerate your Gmail App Password and update `config.properties`
Installer runs but app crashes	Missing JavaFX dependencies in package	Rebuild with `jpackage` using the full

		`target/dependency` folder as `--input`
`NoClassDefFoundError` for JavaFX	Main class extends `Application` directly	Ensure the entry point is `com.brandex.Main`, not `com.brandex.App`

Known Issues & Limitations

The application currently targets **Windows only** for the packaged installer.

Developers

- Malik Bennett
- Ethan Dixon
- Dylan Lee-Sue
- Twyane Campbell